

Major Project / PCSE26 – 29

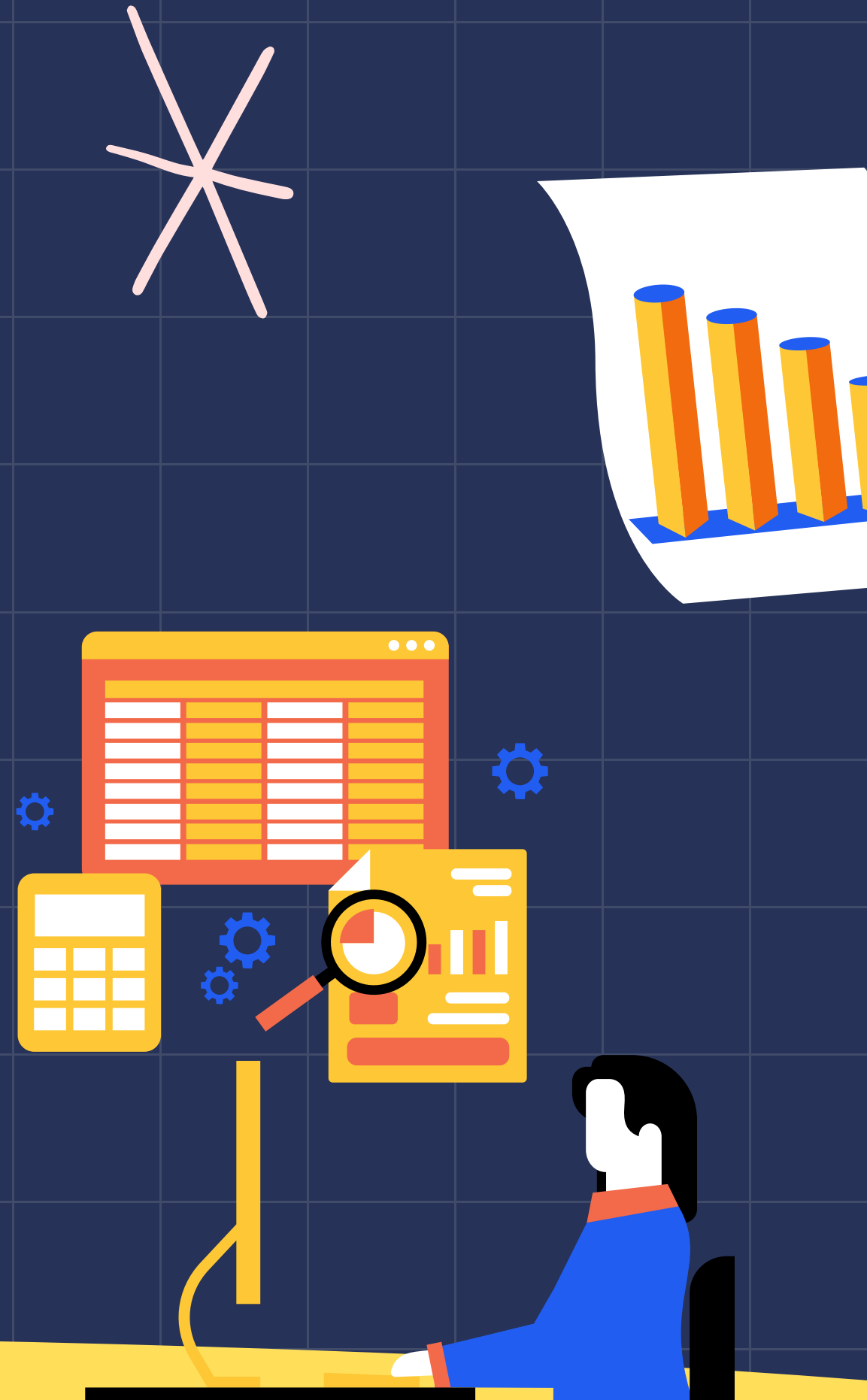
SOCIAL PULSE

Sentiment Analysis Platform

Presented by :

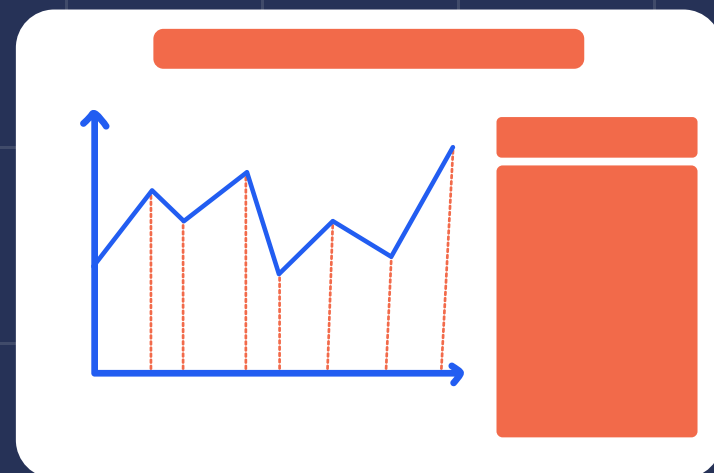
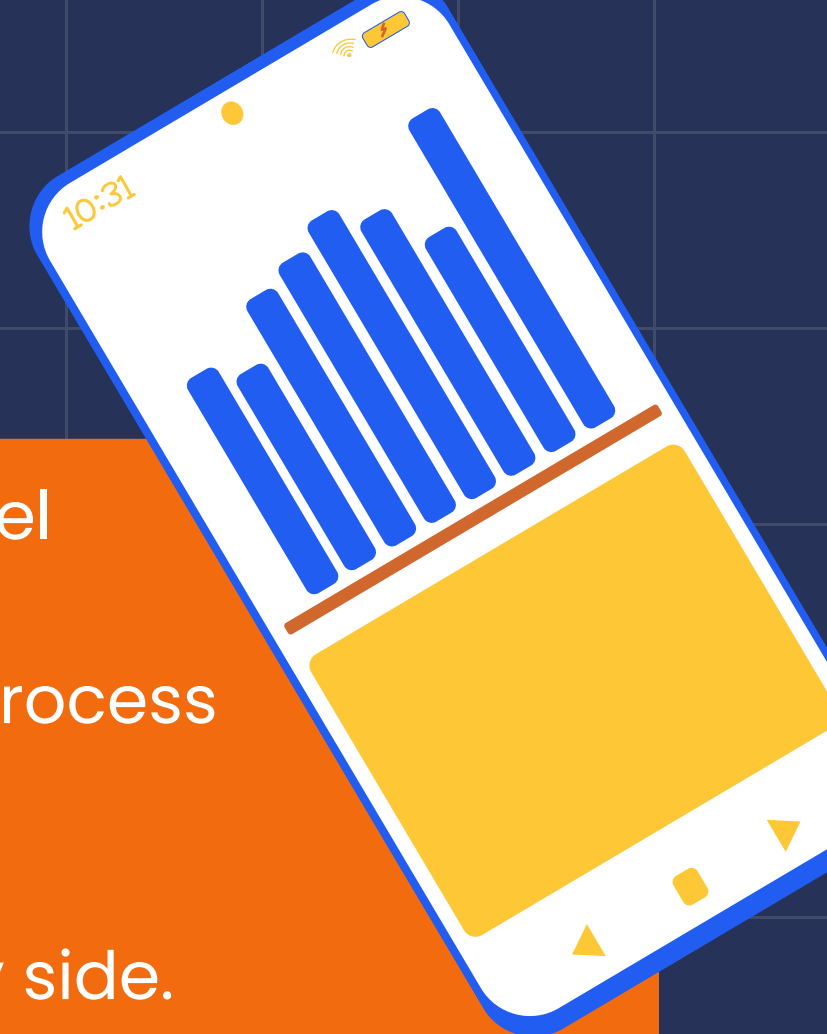
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PROBLEM STATEMENT

- Businesses and small vendors lack real-time insight into what customers actually feel about their products.
- Reviews on Amazon, Flipkart, and social media are unstructured and impossible to process manually at scale.
- Aspect-level feedback is buried inside long, unstructured reviews.
- No unified tool exists to compare sentiment across two competing platforms side by side.
- Decision-making relies on star ratings alone, ignoring the nuance and intent behind written feedback.
- Existing sentiment tools are expensive, generic, and not accessible to small Indian sellers.



PROPOSED SOLUTION

A web-based sentiment analysis platform
for product reviews and brand monitoring

Users can upload CSVs, paste text, or upload
screenshots of reviews from any platform

Analyzes:

-  Sentiment (Positive / Negative / Neutral)
-  Aspect Detection
-  Intent Detection (complaint, praise, inquiry)
-  Emotion Detection (anger, joy, sadness)
-  Platform vs Platform Comparison
-  AI Executive Report (Google Gemini)



REPORT

Developed SocialPulse, a sentiment analysis system for e-commerce product reviews

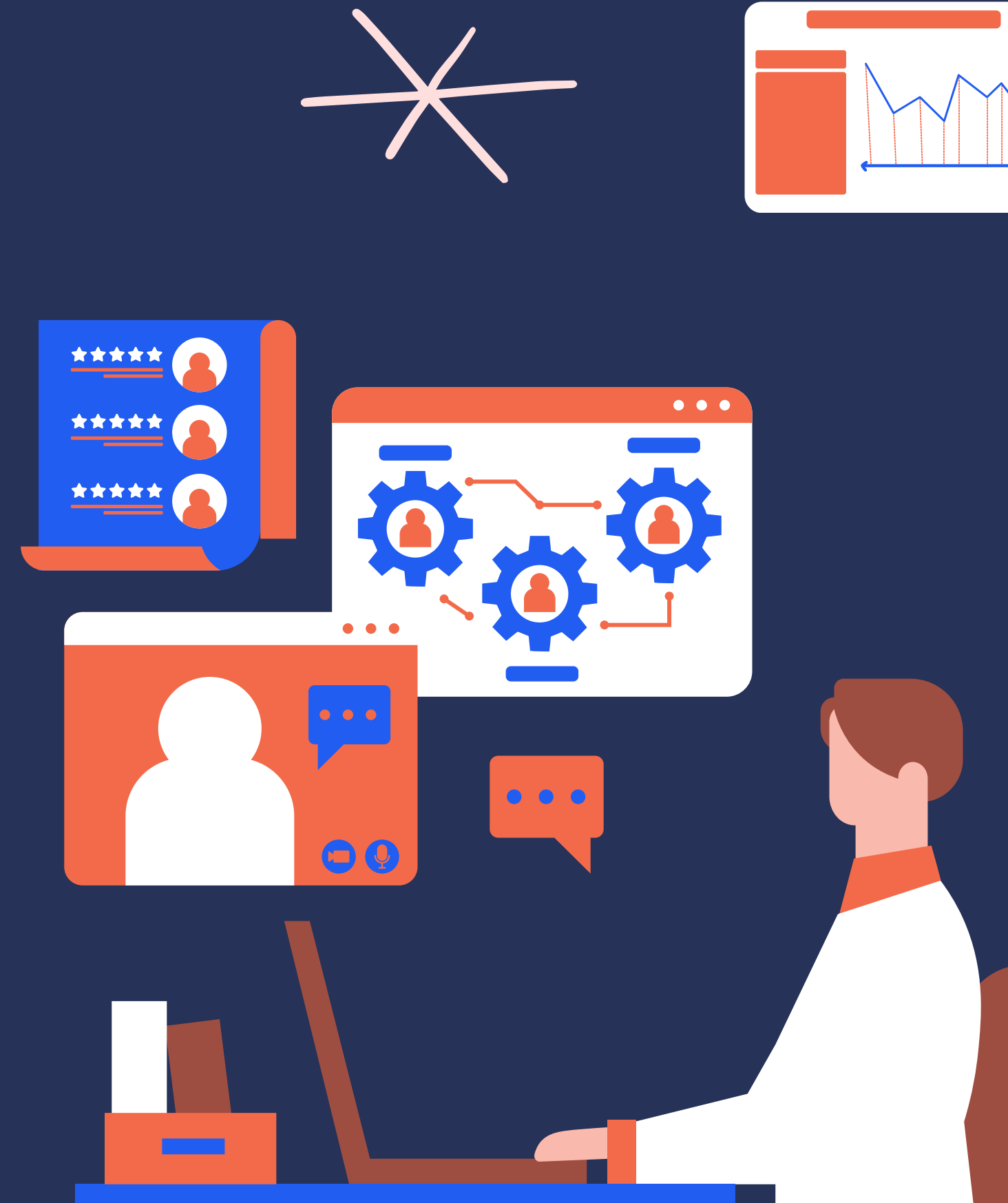
Architecture:

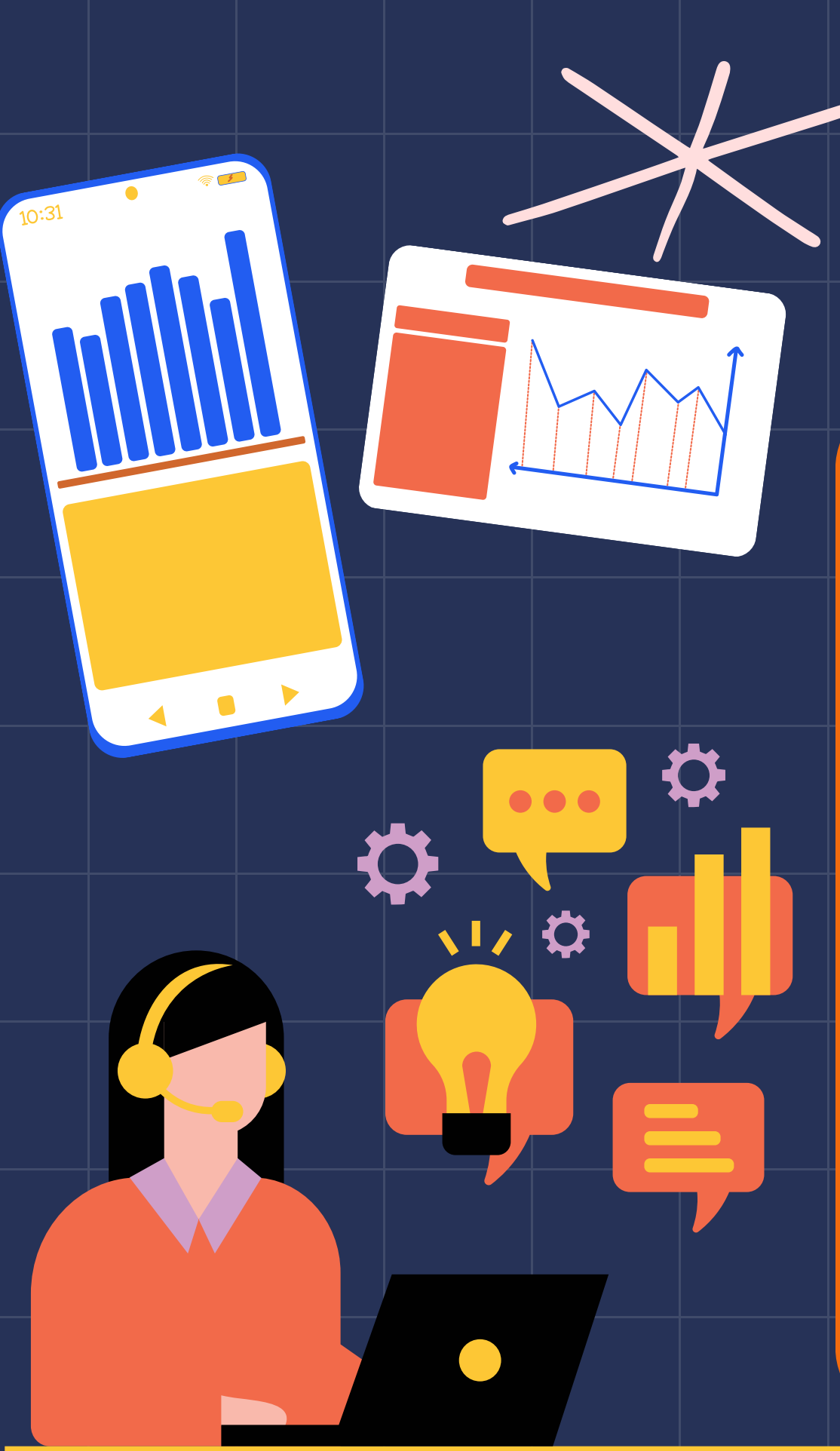
Data Layer → CSV / Text / Image input
Processing Layer. → VADER + Aspect extraction
Presentation Layer → Streamlit dashboard

Methodology:

Text preprocessing + language translation
Rule-based sentiment & aspect scoring
AI-based executive report generation

Provides actionable insights on strengths, weaknesses, and customer pain points





TECHNICAL HIGHLIGHTS & PROJECT DETAILS

Tech stack : Frontend + Backend: Built with Streamlit (Python)

Sentiment Engine: VADER (Valence Aware Dictionary and sEntiment Reasoner) — lexicon-based, interpretable, no training required

Translation: deep-translator (Google Translate API) for multilingual review support

AI Layer: Google Gemini generates executive-ready SWOT reports from aggregated review data

Image OCR: Claude Vision API for extracting reviews from screenshots

Visualization: Altair, Matplotlib, WordCloud

RESEARCH PAPER

Paper titled:

"A Lightweight Multi Domain Sentiment and Aspect Framework Using Python Rules"

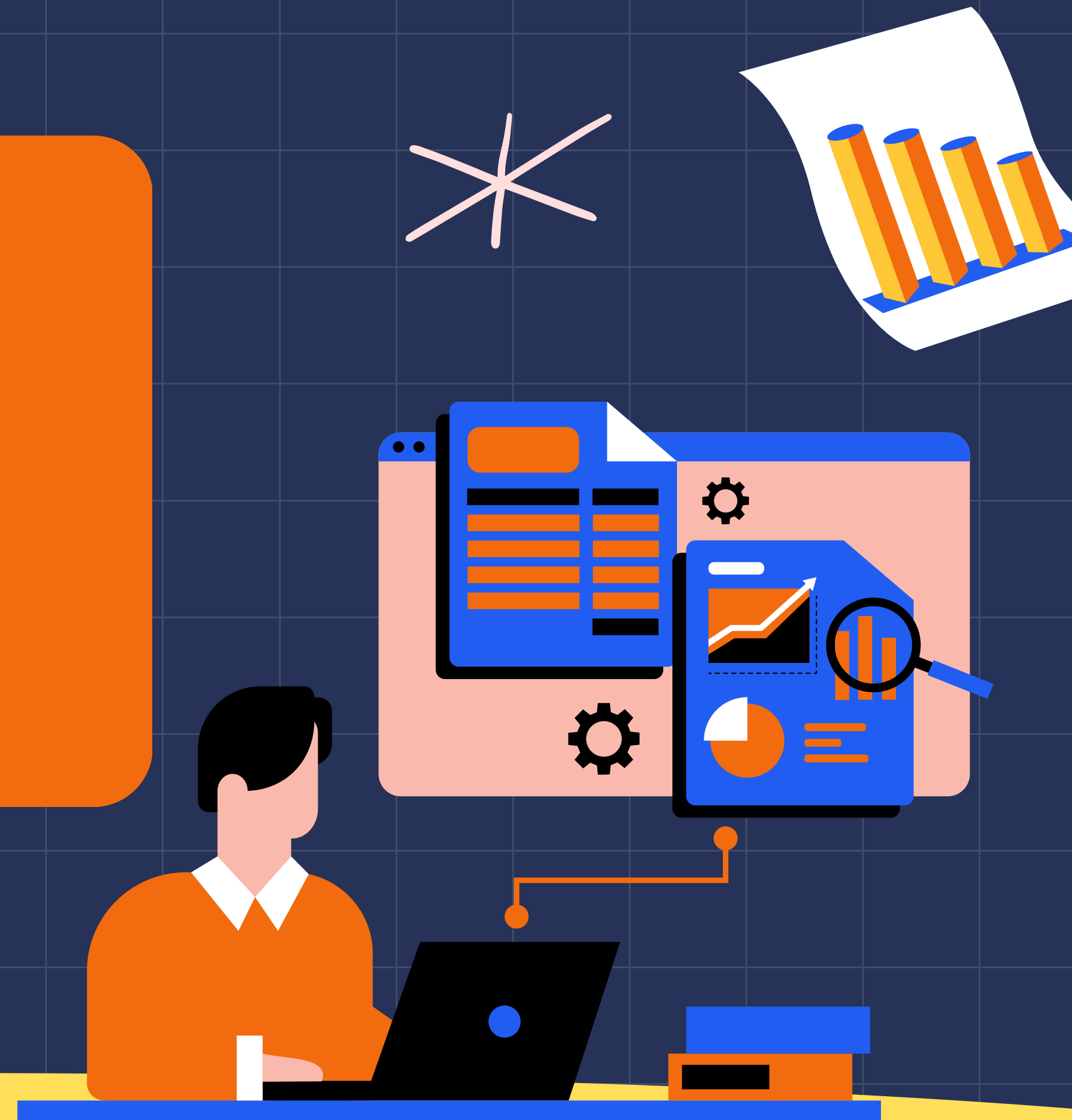
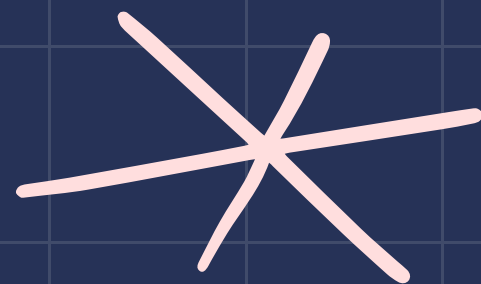
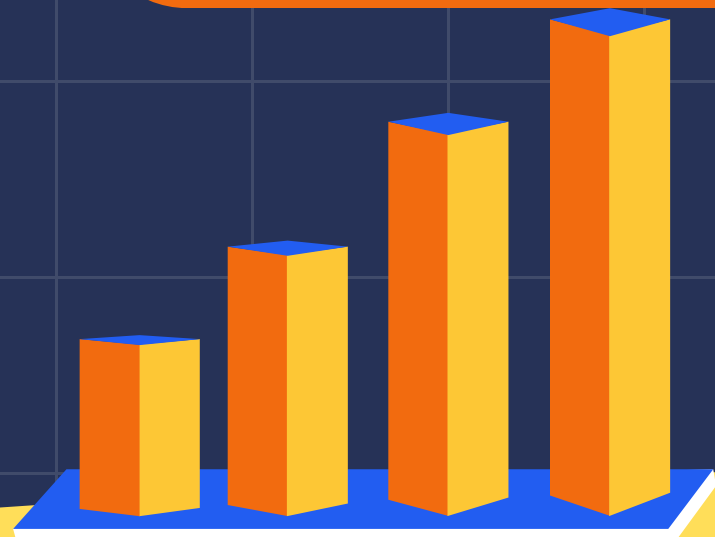
Submitted to:

2nd IEEE International Conference on Advances in Computing, Communication & Networking (ICAC2N 2026)

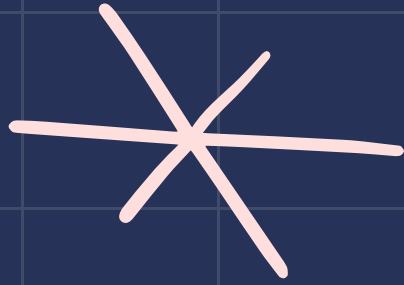
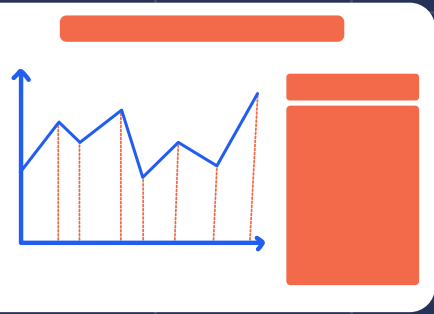
Paper ID: 988

Key contributions:

- Lightweight sentiment system — no GPU required
- VADER lexicon + custom Python rule engine
- Multi-domain aspect detection (electronics, fashion...)
- Deployable on standard consumer hardware
- Accessible to students and small-scale developers



CHALLENGES FACED



- Dependency conflicts between translation library(googletrans) and modern HTTP clients
- Making platform comparison generic (not hardcoded to Specific big Brands)
- Aspect detection returning false negatives → solved with synonym keyword lists
- Balancing speed vs accuracy: VADER is fast but struggles with sarcasm and complex negations

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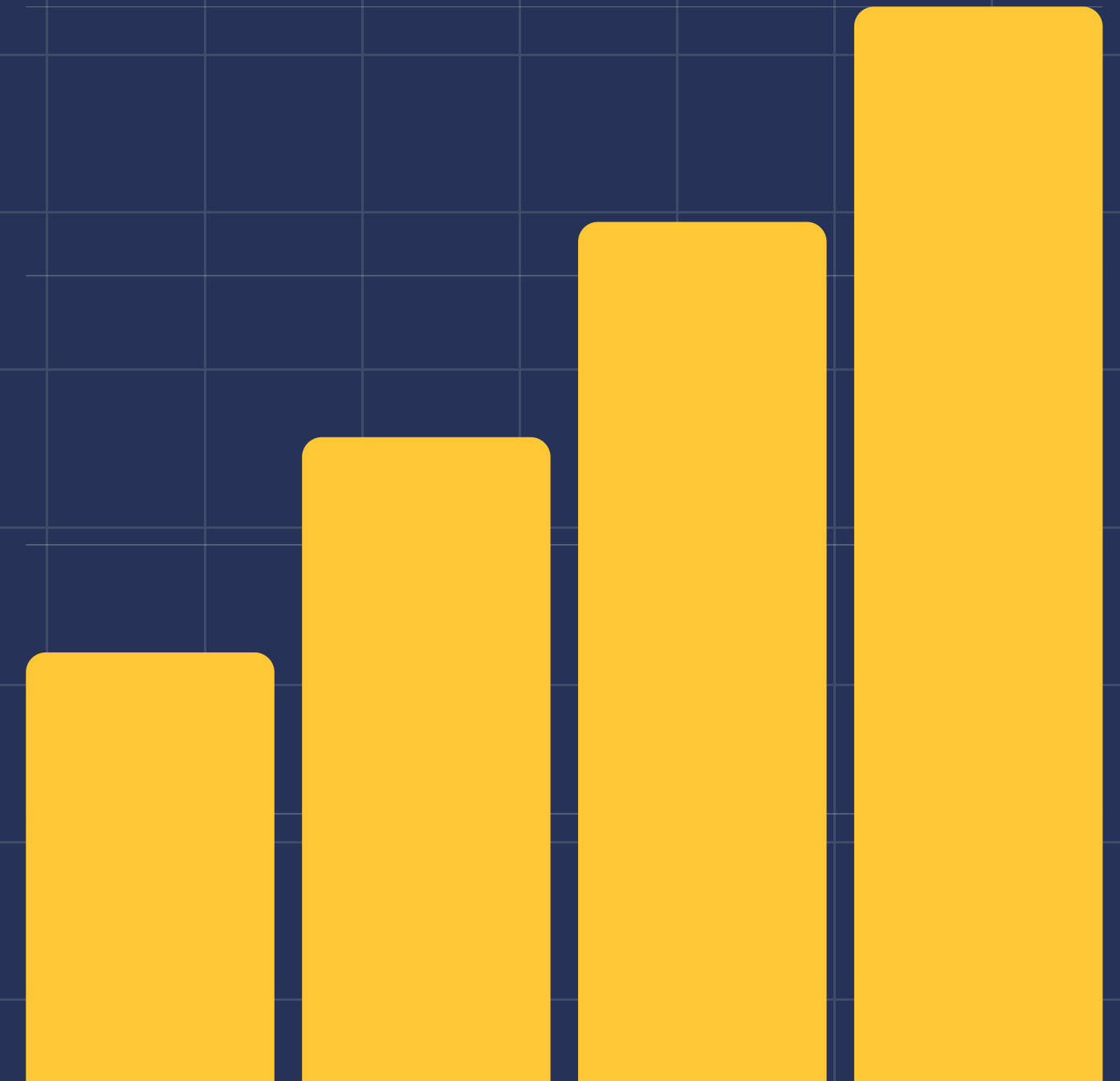
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Item 1

Item 2

Item 3

Item 4





RESULTS / OUTCOMES

- Successfully built a working multi-feature sentiment analysis dashboard
- Real-time single review analysis with aspect, intent, and emotion detection
- Platform comparison across any two sources using CSV, text, or image input
- AI-generated executive report with SWOT analysis and recommended actions
- Smooth end-to-end flow:
Upload → Analyze → Visualize → Report

THANK YOU

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